

COMPARING EXTREME RESPONSE STYLES BETWEEN AGREE-DISAGREE AND ITEM-SPECIFIC SCALES

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Abstract Although Likert scales in agree-disagree (A/D) format are popular in surveys, the data quality yielded by them is controversial among researchers. Recognizing the measurement issues involved with the A/D format, researchers have developed other question formats to measure attitudes. In this study, we focused on an alternative question type, the item-specific (IS) question, which asks the respondent to choose an option that best describes his or her attitude. Using political efficacy items from the American National Election Studies (ANES), we compare extreme response style (ERS) between A/D and IS scales. Latent class factor analysis showed that ERS exists in both A/D and IS scale formats, but differs slightly across the two. Also, when analyzing ERS within subjects across two waves, there is only a single ERS for both question formats, after controlling for the correlation within respondents. The last finding suggests that ERS is a stable characteristic.

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When designing survey questions to measure opinions and attitudes, one of the first ideas that comes to a researcher's mind is probably a Likert scale. As Bradburn, Sudman, and Wansink (2004) point out, attitudes comprise cognitions, evaluations, and behaviors. The cognitive component is one's beliefs about the target object. The evaluative component is one's evaluation of the object. The behavioral component is the connection between attitude and behavior. A Likert scale can be used to measure both the respondent's evaluation (agree vs. disagree) of the rating object and the strength of the evaluation (strongly vs. somewhat). That is, it measures the intensity of one's attitude toward an object (Krosnick and Abelson 1992). Furthermore, it may seem easy to design Likert scales, since researchers have only to create questions describing the objects they want to measure without varying the response options. These advantages make Likert scales very popular among social scientists and marketing researchers. For example, the Marketing Scales Handbook and other similar references provide numerous citations to publications using Likert scales (Bearden and Netemeyer 1999; Bruner, Hensel, and James 2001).

Although they are popular in many fields, the data quality yielded by Likert scales in the agree-disagree (A/D) format is controversial among researchers (for examples, see Clark and Clark 1977; Billiet and McClendon 2000; Fowler and Cosenza 2008; DeVellis 2011; Revilla, Saris, and Krosnick 2013). Some survey methodologists have even argued that "researchers will have more reliable, valid, and interpretable data if they avoid the agree-disagree question form" (Fowler 2008, 105). One of the concerns about the A/D scale is *acquiescent response style bias*, defined as the tendency to choose "agree" or "yes" responses more frequently than other response options (Baumgartner and Steenkamp 2001). Recognizing the measurement issues involved with the A/D format, researchers have developed other question formats to measure attitudes. In this study, we will focus on one of these question types, the item-specific (IS) question, which asks the respondent to choose an option that best describes his or her attitude. Distinct from the A/D format, which offers the same response options for all questions, an IS question presents response options that are specific to the question contents (see appendix A for examples of IS questions).

Efforts have been taken to compare the quality of data from A/D and IS rating scales. Although some research has examined extreme response style (ERS) using the A/D scale, no work has yet examined ERS in the context of the IS scale. ERS, the tendency to select the two endpoints of a response scale more frequently than the intermediate ones (Paulhus 1991), produces another common measurement bias in rating scales. This study fills that gap by examining ERS with A/D and IS scales using experimental data in a panel study. We also answer two research questions. First, do respondents show different patterns of ERS when using an IS scale compared to an A/D scale? Since ERS is regarded as a type of measurement error, a scale that yields lower levels of

ERS bias is preferred. Second, is ERS a stable trait over time or is it a phenomenon mainly driven by survey design features, such as question formats?

Literature Review

The literature presents a number of problems associated with questions in the A/D format.

First of all, to design an effective scale, researchers must make a series of decisions about the format of the question, including the number of response options (Alwin and Krosnick 1991; Revilla, Saris, and Krosnick 2013), whether to label the whole scale or only the endpoints (Krosnick and Berent 1993), whether to use numeric or verbal labels (Alwin and Krosnick 1991), and whether to provide middle options (O'Muircheartaigh, Krosnick, and Helic 2000). These decisions can potentially influence the reliability as well as the validity of the data. (For a more comprehensive review, see Krosnick and Presser [2010].) Second, the choice of the rating object is arbitrary, and different wordings are subject to differential measurement errors (DeVellis 2011). The common practice is to phrase the statement such that it aligns unambiguously with one end of a continuum, although the endpoint of a latent continuum may also be arbitrary (Fowler 1995). Third, the cognitive process of answering an A/D question is different from that used to answer most other survey questions. The cognitive burden is higher than if the attitude is approached more directly through IS format. Rather than the well-known four-step cognitive process for survey response (Tourangeau, Rips, and Rasinski 2000), the A/D format requires a more involved set of cognitive steps (Carpenter and Just 1975; Clark and Clark 1977; Trabasso, Rollins, and Shaughnessy 1971). The fourth problem associated with the A/D format is that it is not clear what the distance is between the two adjacent options (Fowler and Cosenza 2008). In analysis, researchers often dichotomize Likert scale items to agree versus disagree, and hence information is lost from the supposedly ordinal scale (Fowler 2008). Last but not least, A/D format suffers from acquiescent response-style bias (Schuman and Presser 1981; Billiet and McClendon 2000; Billiet and Davidov 2008). Such response style can artificially increase the estimated mean and make the responses look more positive than they really are.

Knowing the measurement errors associated with the A/D scale, some survey methodologists prefer to use IS formats over A/D formats because IS formats are "much simpler, direct and informative" (Fowler 1995, 57; see also Converse and Presser [1986]). In contrast to the A/D format, in the IS format, "the categories used to express the opinion are exactly those answers that we would like to obtain for this item" (Saris et al. 2010). Thus, it is reasonable to assume that the cognitive burden of answering an IS question is lower than that of answering an A/D question. The respondent needs only to judge and map his or her attitude onto the response scale without having to map both his

or her true attitude and the rating object onto the response continuum in order to answer an A/D question.

Empirical studies comparing the measurement quality of A/D and IS formats are relatively limited, and the findings are mixed. Most of the studies examine acquiescent response style and the reliability and validity of the A/D and IS scales. Berkowitz and Wolkon (1964), who were among the first to examine these two types of rating scales, discovered that the reliability is similar across all scale types. Later, Ray (1979) tested a total of six A/D and IS scales, and correlated the responses with external validation data.¹ He reported mixed findings: two types of the A/D scale and one IS scale had higher validity than the other scales. Patient satisfaction rating is another area in which A/D versus IS formats have been studied. Counte (1979) compared patient-satisfaction rating scales and found that the IS scale explained more variance than did the A/D measures. Similarly, Ross, Steward, and Sinacore (1995) evaluated seven patient-satisfaction rating scales, three of which were A/D Likert scales. They detected acquiescence bias for all scale types.² They also found that the internal consistency of satisfaction measures was higher for respondents who gave either no acquiescent responses or a moderate amount, as compared to those with a high level of acquiescent responses. Saris et al. (2010) extended this line of study to an international setting (i.e., 14 European countries) using a multitrait-multimethod approach. They demonstrated that, for two out of three questions, the item-specific format showed better reliability, validity, and quality than the A/D format.³ However, the quality advantage for the IS question was not universal across all countries examined. More recently, Cibelli and Callegaro (2011) conducted a Google AdWords customer satisfaction survey in five languages; in each language, the A/D format elicited more acquiescence bias than did the item-specific format.

Previous research has also shown that the A/D Likert scales are susceptible to ERS bias (Greenleaf 1992; Weijters, Schillewaert, and Geuens 2008; Morren, Gelissen, and Vermunt 2011; Vaerenbergh and Thomas 2012). This is true regardless of the design features of the A/D scale, including scale length (5 to 11 points), middle response category, fully labeled versus end labeled, numerical labels, and agreement scale versus bipolar scale, although the magnitudes are slightly different (Moors 2003; Kieruj and Moors 2010, 2013; Moors, Kieruj, and Vermunt 2014). The presence of extreme response style is stable over time in longitudinal studies and consistent across items in the same survey, suggesting that it is a stable personality trait among respondents

1. The external validation data in the Ray (1979) study are the peer ratings of respondents' achievement motivation. The validity is operationalized as the correlation between the self-reports and the peer ratings.

2. The acquiescent response bias in Ross, Steward, and Sinacore (1995) was measured through pairs of questions for which the content is the same but the direction of the question wording was reversed.

3. Validity in Saris et al. (2010) refers to the internal validity as measured by the MTMM model.

(Weijters, Geuens, and Schillewaert 2010a, 2010b).⁴ It is unknown whether the IS scale also suffers from ERS bias and to what extent compared to the A/D scale. The goal of this study is to evaluate and compare the ERS bias between these two scale formats through experimental data.

Data and Measures

This study analyzes the experimental data embedded in the 2012 American National Election Studies (ANES). The questions under study measure the general population's political efficacy. Political efficacy refers to the "feeling that individual political action does have, or can have, an impact upon the political process" (Campbell, Gurin, and Miller 1954, 187). The ANES has measured the general population's political efficacy for decades using the A/D format. Given the importance of this measure, it is critical to achieve a thorough understanding of the measurement of these questions. In 2012, the ANES includes four A/D items to measure individuals' sense of efficacy. The first two items measure one's internal efficacy, and the last two items measure one's external efficacy. The internal efficacy questions ask about how people think about their ability to adequately understand and effectively participate in politics, and the external efficacy questions ask about people's perceptions of the responsiveness of public officials and government institutions to their demands (Niemi, Craig, and Mattei 1991). Although there are debates about the concept of political efficacy, these are beyond the scope of this study, and readers can find more details elsewhere (Balch 1974; Morrell 2003; Chamberlain 2012).

The 2012 ANES conducted the surveys in two modes, namely face-to-face and web, using two independent national representative samples and one identical questionnaire. The target population for the survey was US citizens aged 18 or older by Election Day 2012. It contained a nationally representative main sample and two oversamples, one for African Americans and one for Hispanic Americans. The face-to-face respondents came from an address-based, stratified, multi-stage cluster sample. The web survey was conducted using the GfK KnowledgePanel, a national probability web panel in the United States. Both address-based sampling and random-digit dialing were used to recruit the panel members from the general population. After a household was selected, within-household enumeration was conducted during the recruitment stage and one person from each household was selected to become the panel member. For those households without Internet access, the survey company provided free

4. The ERS in Greenleaf (1992) is operationalized as the proportion of the extreme answers to 16 questions with low inter-item correlations. The ERS in Weijters, Schillewaert, and Geuens (2008) is measured through representative indicators response style means and covariance structure (RIRSMACS) method. The ERS in Weijters, Geuens, and Schillewaert (2010a) is estimated by the multiple-indicators, multiple-covariates (MIMIC) model. The ERS in Weijters, Geuens, and Schillewaert (2010b) is modeled as a tau-equivalent factor complemented with a time-invariant autoregressive effect.

Internet service and computer in order to remove the coverage bias of the panel. As such, the web panel is representative of the general US population.⁵

The ANES comprised two waves of data collection, including a pre-election study and a post-election study. The face-to-face pre-election study was conducted between September and October 2012, and web pre-election study was conducted between October and November 2012. The post-election study for both modes was conducted between November 2012 and January 2013. The response rates for the pre-election were 38 percent and 2 percent for face-to-face and web, respectively (AAPOR RR1). The re-interview rates for the post-election studies were 94 percent and 93 percent, respectively. In total, the ANES completed 5,914 interviews in the pre-election study (2,054 in face-to-face surveys) and 5,510 interviews in the post-election study (1,929 in face-to-face surveys).

In order to experimentally compare the impact of A/D versus IS formats on data quality, the 2012 ANES included four IS items in addition to the A/D items for measuring political efficacy. The IS items were designed to measure the same internal and external efficacy as the A/D items. The two question formats were tested in a between- and within-subject experiment across two waves of interviews. In the pre-election study, the respondents were randomly assigned to one of the two kinds of scales. As a result, 3,023 respondents answered questions with an A/D scale and 2,890 respondents answered comparable questions with an IS scale. In the post-election study, the assignment was reversed. The between-subject design allows for examination of the ERS between the two question formats if we treat the pre- and post-surveys as two cross-sectional surveys. The within-subject design permits study of the change in response style between different question formats for the same respondent. The experimental design and sample sizes are illustrated in [figure 1](#). To rule out the possibility of differential nonresponse bias between the two modes, we compared the demographic variables, including gender, age, race/ethnicity, education level, household income, and marital status, in the face-to-face and web samples. None of them was statistically different between the two modes (see appendix B). Also, we compared the same set of demographic variables between the two rating scale formats within each mode, and they did not differ significantly either. We applied weights to all analyses throughout the paper. The weights were provided by the survey organization.⁶

Analytical Approaches

BETWEEN-SUBJECT EXPERIMENT ANALYSIS

We first examine the distributions of answers to each question between the two question formats using the pre-election interview data. This will indicate

5. For more information about the GfK KnowledgePanel sample design, see [http://www.knowledgenetworks.com/ganp/docs/KnowledgePanel\(R\)-Design-Summary.pdf](http://www.knowledgenetworks.com/ganp/docs/KnowledgePanel(R)-Design-Summary.pdf).

6. For more information on how the weights were created for ANES, please see http://www.electionstudies.org/studypages/anes_timeseries_2012/anes_timeseries_2012_userguidecodebook.pdf.

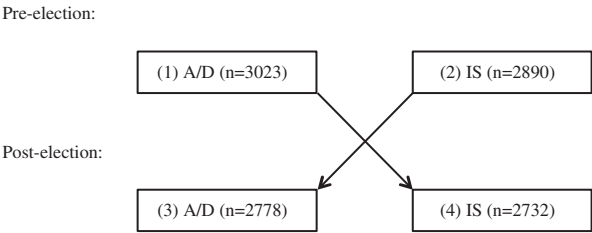


Figure 1. Experimental Design and Sample Sizes.

whether the A/D and IS formats differ in terms of ERS and whether the difference is driven by any particular question or whether it is a consistent phenomenon across all four questions.

We will then perform latent class factor analysis to examine ERS between the two question formats. In the literature, there is no single accepted statistical procedure for measuring the ERS. We choose this method because it can simultaneously estimate response style and the substantive contents of the scale as different latent class factors. As a result, the estimate of the response-style factor is not confounded with the question contents. Billiet and McClendon (2000) initially propose using confirmatory factor analysis (CFA) to estimate both content factors and the acquiescent response-style factor (see also Billiet and Davidov [2008]). CFA treats the rating scales as ordinal variables, so it is adequate for measuring acquiescent response style because a monotonic relationship is assumed between the response-style latent variable and rating scale. In other words, respondents with higher levels of acquiescent response style are more likely to choose a category closer to the positive end than the negative end of the scale. When it comes to the ERS, a U-shaped relationship is expected between the ERS latent class factor and response items. That is, respondents with higher levels of ERS are more likely to choose endpoints than the intermediate categories. CFA cannot capture such a relationship.

Moors (2003) extends this approach to the latent class analysis framework and treats the rating scales as nominal variables in order to estimate ERS. Morren, Gelissen, and Vermunt (2011) further extend the model, imposing an ordinal relationship between the rating scales and content latent class factors but for ERS, maintaining a nominal relationship between the same rating scales and ERS latent class factors in order to capture the non-monotonic (U-shaped) relationship. This is a more parsimonious model; for each response item, only one coefficient is estimated for the content factors. The model was further simplified by imposing an equality constraint on the style factor coefficients (Kieruj and Moors 2013; Moors, Kieruj, and Vermunt 2014). In other words, only one set of coefficients is estimated for all items. This is theoretically meaningful, since the style factor should affect each item equally regardless of the question content. This constraint is similar to the equal factor loading

constraint in the CFA model by Billiet and colleagues (Billiet and McClendon 2000; Billiet and Davidov 2008). The model can be written as follows:

$$\log \frac{P(Y_{ij} = c + 1 | F1_i, F2_i, E_i)}{P(Y_{ij} = c | F1_i, F2_i, E_i)} = (\beta_{0_{jc+1}} - \beta_{0_{jc}}) + \beta_{1_{j_1}} F1_i + \beta_{2_{j_2}} F2_i + (\beta_{3_{jc+1}} - \beta_{3_{jc}}) E_i, \quad (1)$$

where Y_{ij} denotes the response of respondent i to item j , $i=1, \dots, I$, $j=1, \dots, 4$;

$F1_i$ denotes the internal efficacy latent class factor;

$F2_i$ denotes the external efficacy latent class factor;

E_i denotes the extreme response style latent class factor;

$\beta_{1_{j_1}}$ denotes the effects on the adjacent category logits for the internal efficacy latent class factor, $j_1=1, 2$;

$\beta_{2_{j_2}}$ denotes the effects on the adjacent category logits for the external efficacy latent class factor, $j_2=3, 4$;

$\beta_{3_{jc+1}} - \beta_{3_{jc}}$ denotes the nonmonotone (U-shape) relationship between the extreme response-style latent class factor and the items;

and c denotes the number of response options, $c=1, 2, 3$, or 4 .

To study ERS bias for the two types of rating scales, we fit the same measurement model for the four groups (pre-election A/D, pre-election IS, post-election A/D, and post-election IS). As illustrated in figure 2, items 1–4 refer to either the four A/D items or the four IS items, depending on the experimental groups under analysis. The model has two content latent class factors ($F1_i$, $F2_i$) and one ERS latent class factor. The first content factor ($F1_i$) measures internal efficacy, and the second measures external efficacy ($F2_i$). The items can only load on their corresponding content factors. At the same time, all items load on the same ERS latent factor. Previous studies suggest that, given our data structure, this model should fit the data well (Kieruj and Moors 2013; Moors, Kieruj, and Vermunt 2014). We also fit several alternative models in each group to test whether other measurement models can better fit the data. The Bayesian information criterion (BIC) is used to compare models and determine the empirically best-fitting model.

Results

BETWEEN-SUBJECT EXPERIMENT RESULTS

To examine whether ERS is a consistent phenomenon or confined to one or two questions, we first present the proportion of extreme responses for each question from both question formats in the pre-election and post-election surveys (table 1). In both the pre- and post-election surveys, respondents

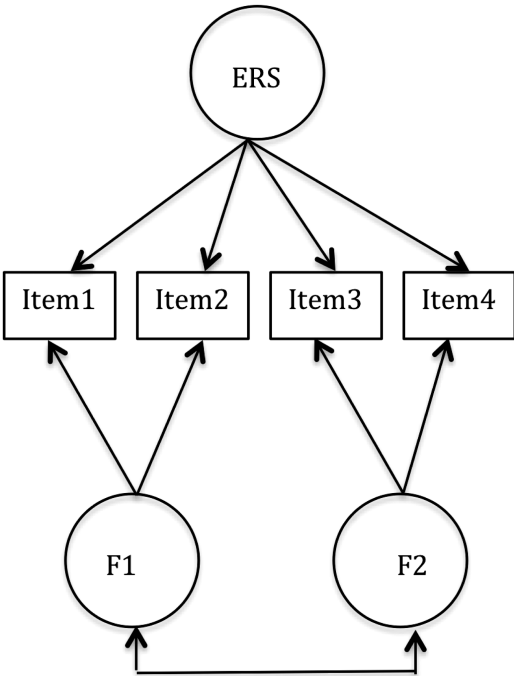


Figure 2. Latent Class Model with One Style Factor and Two Content Factors (items 1–4 represent A/D1–A/D4 or IS1–IS4).

Table 1. Proportions and Standard Errors of Extreme Responses for Agree/Disagree (A/D) and Item-Specific (IS) Scales, 2012 ANES

		A/D		IS		Scale format effect between		
		(1)	S.E.	(2)	S.E.	(1)–(2)	(1)–(4)	(3)–(2)
Pre-election	Complicated	29.00	0.83	15.43	0.67	13.57***	15.29***	15.42***
	Understand	27.79	0.82	14.23	0.65	13.56***	14.91***	9.42***
	Official care	25.77	0.80	22.04	0.77	3.74***	6.77***	3.59***
	Affect							
	government	27.14	0.81	22.34	0.78	4.80***	6.94***	2.27***
		(3)	S.E.	(4)	S.E.	(3)–(4)		
Post-election	Complicated	30.84	0.88	13.70	0.66	17.14***		
	Understand	23.65	0.81	12.88	0.64	10.78***		
	Official care	25.62	0.83	19.00	0.75	6.62***		
	Affect							
	government	24.61	0.82	20.20	0.77	4.41***		

*** $p < .001$

gave more extreme responses to all four questions when using the A/D format compared to the IS format. The independent *t*-test shows that differences in pre-election (i.e., columns (1)–(2)) and post-election (i.e., (3)–(4)) are all statistically significant at $p < .001$. We also compared scale format effects within subject across two waves. The difference between (1) and (4) represents the difference in extreme responses among the same group of respondents who responded using an A/D scale in the pre-election survey and an IS scale in the post-election survey. The paired *t*-test reveals significant differences for all questions at $p < .001$. The comparison between (3) and (2) shows that, for the same respondents, more extreme answers were given using the A/D scale (post-election) than the IS scale (pre-election). The differences are significant at $p < .001$ for the “complicated” and “understand” questions, $p < .05$ for the “official care” question, and marginally significant ($p < .1$) for the “affect government” question. These results offer evidence that the A/D format elicits more extreme answers than the IS format, and that the difference is not confined to a single question or the particular set of respondents but is consistent across all four questions for all respondents.

Although [table 1](#) illustrates that the A/D scale has more extreme responses than the IS scale, it is possible that this difference is confounded with the true attitudinal differences between these two question formats. In other words, it is possible that the difference reflects the actual attitudinal differences the respondent expressed to the two sets of questions, or a combination of the true difference and response-style difference. [Table 1](#) does not tell us to what extent the pattern reflects response option preference, controlling for respondents’ true attitudes toward the content of the questions. In order to control for this potential confounding effect, we performed latent class factor analysis to explicitly control for question content and to separate out the response-style effect from the influence of substantive attitudes. [Table 2](#) shows all the alternative models that we performed and the corresponding BIC. In principle, a model with a small BIC and relatively fewer parameters is preferred. We fit seven different models within each experimental condition for both pre- and post-election surveys. Models 1 and 2 are used to determine the appropriate latent class factors for the data. Model 1 has two content latent class factors only, and model 2 includes the ERS latent class factor in addition to the content latent class factors. For all four experimental conditions, adding the style factor improves the model fit (a smaller BIC), which suggests that the responses, regardless of the question formats, reflect something other than just the content factors. Next, we need to determine the number of classes for the latent class factors by comparing four alternative models, models 2, 3a, 4, and 5, with two, three, four, and five classes, respectively. The BICs decrease monotonically when the number of classes increases from two to four for all four conditions, but the changes are very small when moving from three to four classes. When it comes to five-class models, however, the model fits deteriorate in comparison to four-class models for all but A/D in the pre-election

Table 2. Model Fit Statistics (BIC) for Agree/Disagree (A/D) and Item-Specific (IS) Scales, 2012 ANES

	Pre-election		Post-election		No. of parameters
	A/D	IS	A/D	IS	
Model 1: Content only (2-class)	31410	29185	28146	27203	23
Model 2: Content+ERS (2-class)	30838	28715	27468	26804	40
Model 3a: Content+ERS (3-class)	30483	28643	27119	26757	43
Model 3b: Equality on all latent factors (3-class)	30945	29110	27536	27118	28
Model 3c: Equality on style latent factors (3-class)	30434	28583	27054	26687	31
Model 4: Content+ERS (4-class)	30398	28639	27049	26762	46
Model 5: Content+ERS (5-class)	30381	28654	27069	26785	49

survey. Based on the BIC and ease of interpretation, we use three-class latent class factor models for all four conditions. The model can be further simplified by requiring the β coefficients of ERS to be equal across all items. Models 3b and 3c test two alternative constraints. In model 3b, the equality constraint is imposed on both ERS and the two content latent class factors. In model 3c, only the ERS coefficients are constrained to be equal, but the coefficients associated with the two content factors are allowed to vary across items. The BIC indicates that model 3c fits the data best for both A/S and IS formats in both pre- and post-election surveys. This is theoretically meaningful because the ERS latent class factor is irrelevant to question content, and this is evident by the identical β coefficients across items.

We next report the coefficients of the style latent class factors from model 3c for both pre- and post-election surveys. Figure 3 presents the coefficients for A/D and IS format from the pre-election survey. The coefficients are log odds of the style latent class factor effect on response items. Since model 3c imposes an equality constraint on the style latent class factor, there is only one set of coefficients for each question format. Also, because the response item is treated as a nominal variable, there is one coefficient for each response category. A positive coefficient in figure 3 indicates a positive association between use of ERS and the likelihood of choosing the corresponding response category. As can be seen from figure 3, a common feature shared by the two question formats is that the two endpoints have positive coefficients while the three middle response options have negative coefficients. Recall that the latent class factors are ordinal discrete variables with equidistant category scores. This suggests that a higher level of the style latent class variable represents tendencies toward ERS and a lower level represents a tendency to avoid ERS. However, there are two distinct patterns of ERS associated with these two

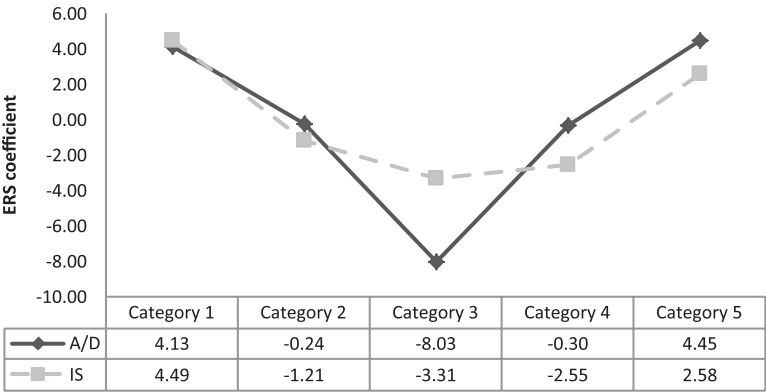


Figure 3. Extreme Response Style Coefficients on Agree/Disagree (A/D) and Item-Specific (IS) Scales, 2012 ANES Pre-Election Survey (see appendix for response categories 1–5).

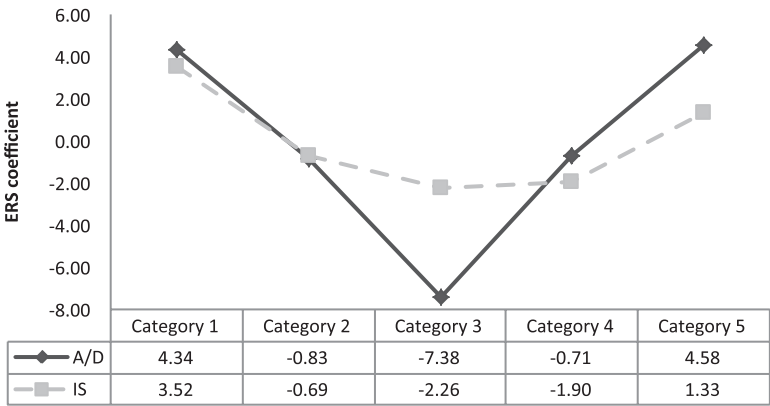


Figure 4. Extreme Response-Style Coefficients on Agree/Disagree (A/D) and Item-Specific (IS) Scales, 2012 ANES Post-Election Survey (see appendix for response categories 1–5).

question formats. For the A/D format, the “agree somewhat” and “disagree somewhat” categories fall between the two endpoints and the middle point, suggesting that the ERS latent class factor reflects the contrast between endpoints and midpoint. For the IS format, the three intermediate categories are close to each other while the endpoints are farther away. It indicates that the ERS coefficient for the IS format reflects the contrast between a preference for extreme responses and a preference for non-extreme responses. Regardless of these question-format specific ERS effects on response categories, the overall effect of ERS on response items is clearly observed from both types of questions. Figure 4 shows the ERS coefficients of A/D and IS formats from the

post-election survey. We find exactly the same pattern as that observed in the pre-election survey. The consistent findings across two waves of surveys lend strong support to the idea that there are two different ERS patterns associated with A/D and IS formats.

WITHIN-SUBJECT EXPERIMENT ANALYSIS

The experimental design embedded in a longitudinal survey not only allows us to test the ERS of two question formats between subjects, but it also permits us to examine within-subject differences between the question formats. Unlike the between-subject analysis, the within-subject analysis must take within-subject correlation into consideration when building the analysis model, since the same respondents answer each question twice: once in A/D format and once in IS format. Figures 5(a) and (b) illustrate two alternative models for examining ERS across two waves. There are two groups of respondents for the within-subject analysis. The first group answered A/D formatted questions in the pre-election survey and IS formatted questions in the post-election survey, and the order of question formats is reversed for the second group. We fit latent class factor models separately for the two groups in order to measure ERS. We are interested in comparing the ERS latent class factors between the two question formats within each group rather than between the two groups. As can be seen from figure 5(a), the A/D questions were asked in the pre-election (or post-election) and IS were asked in the post-election (or pre-election) survey. F_1 and F_2 are the two content latent class factors for the A/D format, and F_3 and F_4 are the two content latent class factors for the IS format. We allow correlation between the four latent class factors because in theory the within-subjects political efficacy should be correlated across question formats and across pre- and post-election surveys. In contrast, there is only one style latent class factor (ERS) for both question formats. This model suggests that within subjects, ERS is stable even though the question formats changed and the questions were administered at two points in time. The model in figure 5(b) contains the same response items and content latent class factors. The only difference is that there are two style latent class factors (ERS_1 and ERS_2) for A/D and IS formats, respectively. The two style factors are correlated with each other. This model suggests that the two question formats elicit two types of ERS within subjects across two time points.

WITHIN-SUBJECT EXPERIMENT RESULTS

Figure 5 presents two theoretical measurement models measuring ERS across time using the panel data. However, we also test alternative models that are slight modifications of the ones in figure 5. The BIC and number of parameters in these models are presented in table 3. Model 1 contains four content latent class factors only, model 2 contains four content factors and one style factor (figure 5a), and model 3 contains four content factors and two style factors (figure 5b). For group 1 members who were asked questions using the A/D

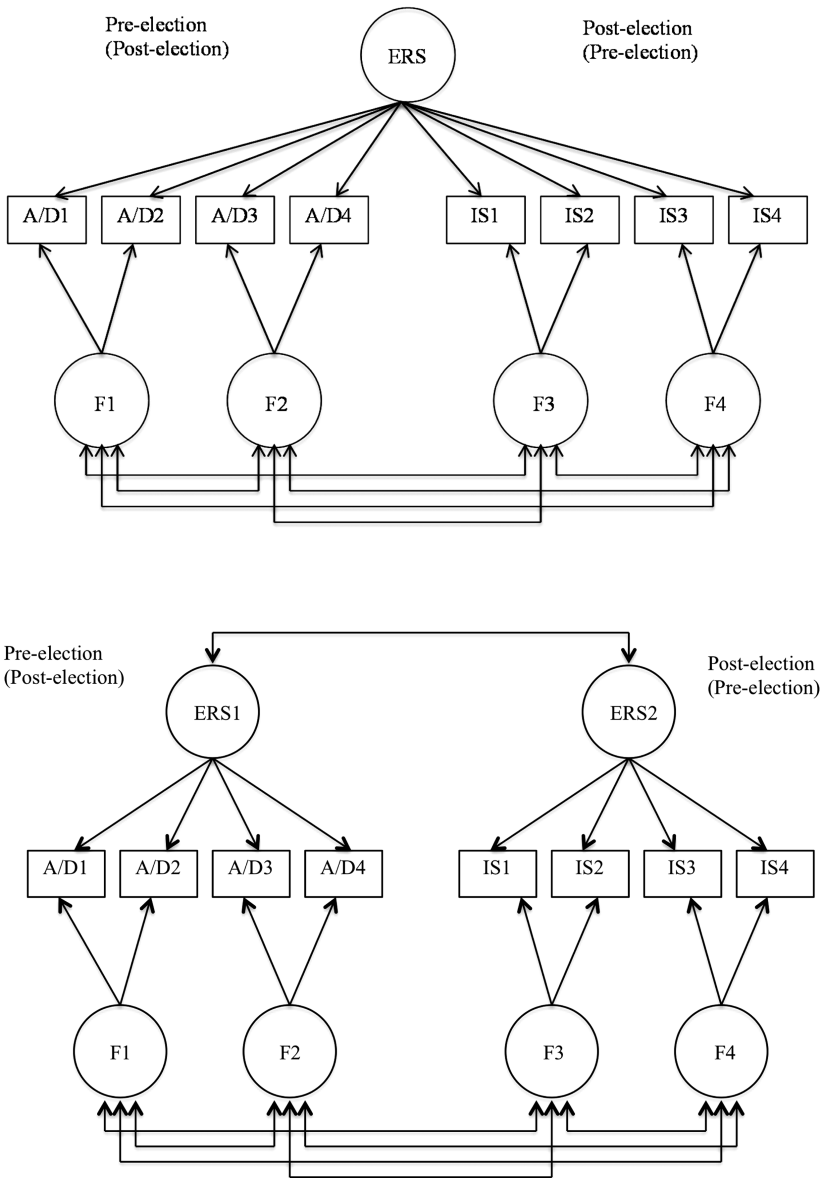


Figure 5. (a) Latent Class Model with One Style Factor and Four Content Factors; (b) Latent Class Model with Two Style Factors and Four Content Factors.

Table 3. Model Fit Statistics (BIC) for Within-Subject Analysis, 2012 ANES^c

	Group 1	Group 2	No. of parameters
	A/D→IS ^a	IS→A/D ^b	
Model 1: Content only	19334	18386	54
Model 2: Content+1 ERS	19201	18214	88
Model 2a: Equality on all latent factors	19619	18570	56
Model 2b: Equality on style latent factor	19144	18091	60
Model 3: Content+2 ERS	19207	18192	91
Model 3a: Equality on all latent factors	19637	18629	71
Model 3b: Equality on style latent factors	19151	18105	67

^aAgree/disagree in pre-election and item specific in post-election survey.
^bItem specific in pre-election and agree/disagree in post-election survey.
^cAll latent class factors have 3 classes.

format in the pre-election and IS format in the post-election, model 2 has the smallest BIC. We then proceed to test two alternatives of this model by imposing equality constraints on the all latent class factors (model 2a) or on the style latent class factor only (model 2b). The second approach results in the smallest BIC, which suggests that for group 1, the model in [figure 5a](#) fits the data best. Similarly, for group 2 members, who received IS questions in the pre-election and A/D questions in the post-election survey, the smallest BIC comes from model 2b as well. This means that the model in [figure 5a](#) consistently outperforms [figure 5b](#), regardless of the question administration order. In other words, the single ERS latent class factor for group 1 indicates that once the within-subject correlation is built in to model, the question formats do not affect ERS.

The ERS coefficients for groups 1 and 2 are plotted in [figures 6](#) and [7](#). The general conclusion from these two figures is that ERS represents a contrast between two extreme categories and the middle three categories. This is evident by the negative and relatively similar coefficients for the three non-extreme categories and the two positive coefficients at the endpoints. The ERS coefficients from the two figures are very similar to each other, which indicates that the single ERS model is quite reliable.

Discussion

In this study, we examined extreme response styles from two types of rating scales, namely agree-disagree (A/D) scales and item-specific (IS) scales, using data from between- and within-subject experiments. The four items we examined come from the 2012 American National Election Studies and measure internal and external political efficacy. The experiments were carried out in two waves of data collection, with respondents randomly assigned to one of the two

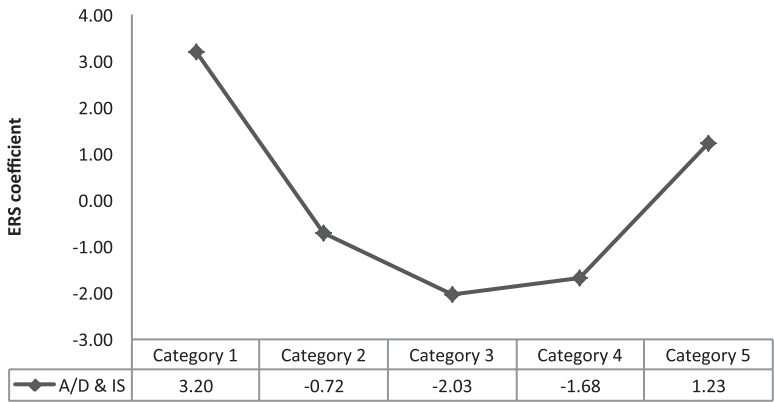


Figure 6. Extreme Response-Style Coefficients for Group 1 (A/D in the pre-election and IS in the post-election), Model 2b (table 3), 2012 ANES (see appendix for response categories 1–5).

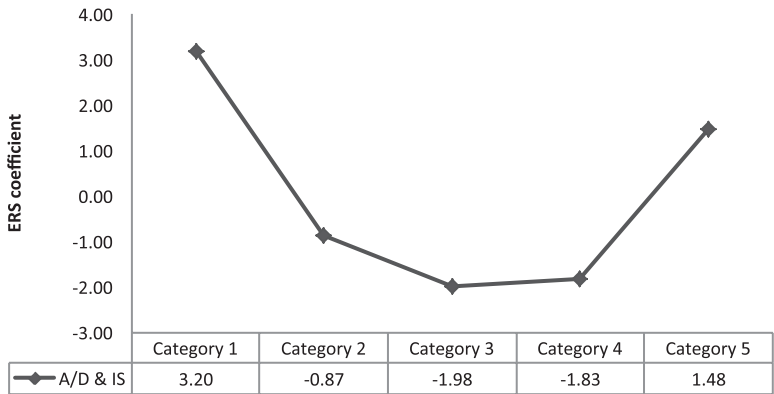


Figure 7. Extreme Response-Style Coefficients for Group 2 (IS in the pre-election and A/D in the post-election), Model 2b (table 3), 2012 ANES (see appendix for response categories 1–5).

question formats in wave 1, and then assigned to the remaining format in wave 2. We conducted a latent class factor analysis that yielded three main findings: first, ERS exists in both A/D and IS scale formats. Ideally, we would have identified a scale format that is response-style free. Unfortunately, this study shows that we cannot achieve this by using item-specific scales. Second, the ERS from the two formats show slightly different shapes. Although respondents who exhibit higher levels of ERS are more likely to choose the two endpoints in both formats, those who exhibit lower levels of ERS show different patterns in these

two formats. In A/D format, ERS reflects a contrast between the two endpoints and the middle category while the two intermediate categories (“agree somewhat” and “disagree somewhat”) are somewhere in between. In IS format, ERS is the contrast between the two endpoints and the three non-extreme categories. This finding is consistent in both pre-election and post-election studies. Third, when analyzing ERS within subjects across two waves, there is only one single ERS for both question formats, regardless of which question type was asked first, after controlling for the correlation within respondents.

Why do the ERSs differ between A/D and IS, and why do they differ in this particular pattern? A closer look at the response options of the two scale formats may shed some light on this question. The verbal labels of these two scale formats are different. The A/D format is a bipolar scale, while the verbal labels for IS seem to be more unipolar. In a bipolar scale, the two endpoints represent the ends of an underlying continuum while the midpoint represents the conceptual center of this continuum. The conceptual distance from “agree strongly” to “neither agree nor disagree” and the distance from “disagree strongly” to “neither agree nor disagree” are equal. Since the respondents can easily pinpoint the conceptual center of an A/D scale, those who tend to avoid extreme points will move away from the endpoints toward the midpoint, the one that is the furthest from the extreme points. Thus, we observe a contrast between the extreme points and midpoints for the A/D scale.

For the IS scale, however, the conceptual center is more difficult to establish. For example, consider the following IS scale: *Extremely well, very well, moderately well, slightly well, or not well at all*. It is clear that “moderately well” is the closest to the conceptual center of the scale but is not necessarily the center. Also, the conceptual distance between “extremely well” and “moderately well” and the distance between “not well at all” and “moderately well” are not necessarily equal. Consequently, when respondents want to avoid extreme options, they will pick one of the middle three categories with relatively equal likelihood, since the verbal labels of response categories on an IS scale do not necessarily contain a conceptual midpoint. These findings are consistent with the results from Moors, Kieruj, and Vermunt (2014). In their study, the balanced bipolar A/D scale with numerical labels from -3 to +3 elicits an ERS similar to the one for the A/D scale in the present study, while a unipolar A/D scale with numerical labels from 1 to 7 has an ERS similar to the one from the IS scale in the present study.

Another important finding is the similarity of the ERS between two question formats within subjects. Each respondent answered both types of scales across two waves of interviews. Our analysis shows that once the within-person correlation is incorporated into the latent class factor analysis, the ERS differences between question formats disappear for both the group of respondents who answered A/D in pre-election and IS in post-election interviews, and the group of respondents who answered IS in pre-election and A/D in post-election interviews. As a result, there is only one ERS latent class factor for both question formats in both groups. This suggests that the response style is stable across two

question formats and across two time points (pre- and post-election). This finding lends strong support to the argument that response styles have more to do with individual respondents than with survey design or question content. This finding is in line with the results reported in previous studies (Kieruj and Moors 2013; Weijters, Geuens, and Schillewaert 2010a, 2010b). Those studies report stable response style for the A/D format in longitudinal surveys or demonstrate an association between response style and personality measures. In the 2012 ANES, the two waves are about four months apart, and the question formats are different in the two interviews. The identical and similar ERS we observed in the data provide further support for the argument that response style is a stable behavior.

One of the aims of this study is to identify a rating scale format that is ERS free. As we already know, A/D format suffers from both acquiescent and extreme response style while IS format reduces acquiescent response bias. This study shows that, like A/D format, IS format is still susceptible to ERS. Thus, the goal of eliminating ERS cannot be achieved by using IS format. However, given the advantage of IS format in reducing acquiescent response style, we would still recommend IS over A/D rating scales, in general. On the bright side, although we fail to remove ERS through scale design, we can still control for ERS, as well as other response styles, in the statistical analysis stage. Latent class factor analysis is one analytic approach that measures substantive latent class factors and response style latent class factors at the same time. The estimates for the substantive latent construct should be free of the response-style bias. Of course, this analytic approach imposes requirements on the rating scales. At the very least, the latent construct needs to be measured by multi-item scales rather than by one single question. Also, reverse-worded questions must be included for measuring acquiescent response style.

As this study shows, more work is required to examine and ultimately eliminate response styles from rating scales. In the present study, we used latent class factor analysis to estimate the response style. As we mentioned above, this is certainly not the only analytical approach one can take. There are other methods in the literature that have been used to examine response style, such as item response theory (Jong et al. 2008), multidimensional unfolding model (Javaras and Ripley 2007), and representative indicators response style means and covariance structure (Thomas, Abts, and Weyden 2014). However, we are not aware of any studies comparing these different analytical tools in this domain of study. Future research should evaluate and compare the existing analytic methods and resolve any differences in the conclusions in order to better adjust for response styles and produce more accurate survey estimates.

Furthermore, although the IS format fails to eliminate ERS, it is possible that other designs of the IS format may elicit different levels of ERS. Future research should explore the scale length and the use of verbal and numeric labeling in IS format. Some research has examined variations of A/D format and their impacts on ERS (Kieruj and Moors 2010; Kieruj and Moors 2013;

Moors, Kieruj, and Vermunt 2014), but this is lacking for IS format. In addition, would these findings change if the study had been conducted in another culture? Numerous studies have shown response-style differences between racial/ethnic groups within country and across different countries (for a comprehensive review, see Yang et al. [2010]; Liu [2015]). As international comparative studies become more prominent, it is critical to identify a scale format that can reduce response-style bias across different countries and cultural groups. So far, there is very little knowledge on this topic. Last but not least, the current study only examined scales measuring political efficacy. We hope future studies will replicate this study by examining scales measuring other topics in order to help ensure that the findings reported here are broadly applicable.

Appendix A. Question Wordings and Response Options for Agree-Disagree and Item-Specific Questions

LIKERT SCALE

A/D1: Sometimes, politics and government seem so complicated that a person like me can't really understand what's going on.

A/D2: I feel that I have a pretty good understanding of the important political issues facing our country.

A/D3: Public officials don't care much what people like me think.

A/D4: People like me don't have any say about what the government does.

Response options to A/D1–A/D4 are agree strongly (category 1), agree somewhat (category 2), neither agree nor disagree (category 3), disagree somewhat (category 4), or disagree strongly (category 5).

ITEM-SPECIFIC ITEMS

IS1: How often do politics and governments seem so complicated that you can't really understand what's going on?

Always (category 1), most of the time (category 2), about half the time (category 3), some of the time (category 4), or never (category 5)

IS2: How well do you understand the important political issues facing our country?

Extremely well (category 1), very well (category 2), moderately well (category 3), slightly well (category 4), or not well at all (category 5)

IS3: How much do public officials care what people like you think?

A great deal (category 1), a lot (category 2), a moderate amount (category 3), a little (category 4), or not at all (category 5)

IS4: How much can people like you affect what the government does?

A great deal (category 1), a lot (category 2), a moderate amount (category 3), a little (category 4), or not at all (category 5)

Appendix B. Means and Standard Errors of Demographic Variables for Agree/Disagree (A/D) and Item-Specific (IS) Scales by Mode of Data Collection, 2012 ANES

	Face-to-face				Web				All			
	A/D	IS	χ^2	p	A/D	IS	χ^2	p	FTF	Web	χ^2	p
Female	51.6	52.5	0.09	0.77	51.9	52.3	0.04	0.85	52.0	52.1	0	0.96
Age												
< 30	21.4	20.9	2.59	0.76	20.1	22.0	4.93	0.42	21.2	21.0	4.81	0.44
30–39	16.0	15.1			15.3	14.6			15.6	15.0		
40–49	18.1	16.4			17.8	15.6			17.3	16.7		
50–59	17.7	20.0			20.7	19.5			18.9	20.1		
60–69	14.9	13.7			15.8	16.7			14.3	16.3		
>= 70	11.7	13.9			10.2	11.7			12.8	10.9		
Race/ethnicity												
White	71.4	70.3	2.13	0.55	70.7	71.1	0.93	0.82	70.9	70.9	0.34	0.95
Black	12.6	11.3			11.8	12.1			11.9	11.9		
Hispanic	10.4	11.4			11.8	10.6			10.9	11.3		
Other	5.6	7.0			5.6	6.2			6.3	5.9		
Education												
High school or less	41.1	40.1	0.70	0.70	38.3	42.0	3.26	0.20	40.6	40.1	0.07	0.96
Some post-high school	29.0	31.3			31.2	29.6			30.1	30.4		
Bachelor's and above	29.9	28.6			30.5	28.4			29.3	29.5		
Household income (US dollars)												
0–49,999	51.4	48.8	0.67	0.88	48.3	50.7	2.60	0.46	50.1	49.5	2.59	0.46
50,000–99,999	28.8	31.1			32.9	30.9			30.0	31.9		

Continued

Appendix B. Continued

	Face-to-face				Web				All			
	A/D	IS	χ^2	p	A/D	IS	χ^2	p	FTF	Web	χ^2	P
100,000–149,999	11.6	11.7			11.4	12.0			11.7	11.7		
150,000+	8.2	8.4			7.4	6.3			8.3	6.9		
Marital status												
Married	51.6	54.9	7.36	0.12	55.3	51.2	6.05	0.20	53.2	53.3	0.62	0.96
Widowed	6.2	5.9			4.9	6.5			6.1	5.7		
Divorced	11.9	13.3			12.8	13.3			12.6	13.1		
Separated	3.6	1.6			2.0	2.6			2.6	2.3		
Never married	26.7	24.4			25.1	26.3			25.5	25.7		

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